

Excess nitrogen may decrease maize grain yields when water is limited

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ABSTRACT

Water limitation and nitrogen (N) availability interact antagonistically to constrain maize productivity, yet this critical relationship remains poorly quantified in semi-arid cropping systems. Using maize (*Zea mays* L.) as a model crop, we conducted a three-year (2021–2023) field experiment in the US Great Plains with two irrigation regimes (100% and 70% of plant evapotranspiration targets) and six N availability levels (50–327 kg N ha⁻¹, including fertilizer, irrigation, and residual soil inorganic N). Under full irrigation, maize grain yield and aboveground biomass increased with N availability. In contrast, excess N was detrimental under water-limited conditions, reducing grain yield, aboveground biomass, and root biomass relative to moderate N rates. Maximum grain yield under water limitation required, on average, 31% less N than under full water. Water limitation decreased maximum yields by 36%, 59%, and 30% in 2021, 2022, and 2023, respectively. Notably, yield reductions at high N availability persisted in 2023 despite lower fertilizer inputs due to residual soil N carryover. Nitrogen uptake increased with N supply and was higher under full irrigation; however, under water stress, elevated N uptake became decoupled from yield and biomass production. These results demonstrate that matching N inputs to water-limited yield potential is essential to maximize grain yield and resource-use efficiency. Adjusting N inputs based on soil N availability and yield potential will become increasingly critical as water scarcity intensifies in semi-arid cropping systems.

1. Introduction

Water is often the most limiting resources for crop growth with increased water limitations potentially becoming more common in the Great Plains Region (Deines et al., 2020; Derner et al., 2015; Plett et al., 2020). This threatens the viability of rural economies that rely on irrigation as well as food security as irrigated lands are disproportionately productive; accounting for roughly 20% of arable land yet responsible for nearly 40% of calorie production worldwide (Garces-Restrepo et al., 2007). Given this, preserving water resources and improving water use efficiency is gaining attention as water limitations become more widespread and frequent across the globe (Nozari et al., 2024; Schipanski et al., 2023; Zou et al., 2021). As the global population grows, agricultural demand grows, and under *status quo* projections, the demand for agricultural products will increase by 75% by 2050 compared to 2005 levels (Godfray et al., 2010; Hunter et al., 2017; Tilman et al., 2011). Therefore, it is important to develop productive cropping systems

adapted to limited water conditions while also protecting water quality and availability for future generations.

Despite extensive research, the interaction between nitrogen (N) availability and crop performance under water limitation remains unresolved, with conflicting evidence on whether increasing N alleviates or exacerbates water stress. When water is limited, crops can be co-limited by N due to reduced soil N supply and mobility (Borken and Matzner, 2009; He and Dijkstra, 2014), hindering crop N uptake (Gonzalez-Dugo et al., 2010) and reducing grain yield. Previous studies have suggested that adding more N can help crops overcome this N co-limitation with water by increasing N availability and N uptake, resulting in increased yield and water use efficiency under water limitations (Di Paolo and Rinaldi, 2008; Li et al., 2019; Rudnick and Irmak, 2013). These yield increases can be attributed to additional N increasing photoprotective mechanisms (Ahmadi et al., 2010; Eissa and Roshdy, 2019; Song et al., 2019), osmotic adjustment and cell membrane stability (Nematpour et al., 2019; Nematpour and Eshghizadeh, 2024; Zhang et al., 2012),

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root growth, root length, and water use (Liu et al., 2018; Ogola et al., 2002; Wang et al., 2018), sink potential and grain fill (Ullah et al., 2022), and stomatal conductance and photosynthetic rates (Dinh et al., 2017; Liao et al., 2022; Zhong et al., 2019). These findings implicitly assume that N and water are co-limited and that increasing N availability will improve crop performance under water stress.

However, this assumption has not been consistently supported and there is conflicting evidence about the positive effects of high N with low water availability. Some studies have found that in low yielding, water limited environments, additional N availability has no yield response (Obour et al., 2022), while others have reported that too much N availability can be detrimental to plant growth when water is limited (Chen et al., 2024; Flynn et al., 2023; Pandey, Maranville, and Admou, 2000). Taken together, these findings suggest that N and water may not always be co-limiting and that excess N availability under limited water conditions may reduce yield. Additionally, water limitations can reduce crop N demand and uptake even when N availability is high (Gonzalez-Dugo et al., 2010; Pandey, Maranville, and Chetima, 2000; Peng et al., 2010), thus N co-limitations may not be overcome. Along with potential yield reductions, excess N can hurt ecosystem services such as reducing soil carbon relative to “optimal N” (Poffenbarger et al., 2017; Singh, 2018), and cause soil acidification (Tian and Niu, 2015). Reduced plant N uptake can increase nitrate (NO_3) leaching by as much as 34% due to elevated residual soil N concentrations (Loecke et al., 2017). Losses of N to the environment are already high and increase rapidly when excess N is applied (Puntel et al., 2016). Agricultural N losses have contributed to nearly doubling the amount of reactive N in the biosphere during the last century (Battye et al., 2017; Gruber and Galloway, 2008), with use and demand for N expected to increase further in the agricultural sector in the future (Vishwakarma et al., 2022; Zhang et al., 2015). Therefore, there is urgent need for a better understanding of how N and water availabilities affect crop growth to help evaluate the tradeoffs associated with N use. The discrepancies in the literature could be due to several reasons including the duration of the study, field studies compared to controlled environments, and limited range of N and water treatments. Therefore, multi-year field studies with a broad range of N treatments under contrasting water regimes are needed to better understand how N and water interactions influence crop performance.

Effectively managing N is challenging. Producing high cereal crop yields requires high total N uptake, but there is often asynchrony between soil N supply and crop N demands over the course of the season (Donovan et al., 2025; Grandy et al., 2024; Zhu et al., 2025). Grain yields and soil-crop dynamics are hard to predict, making it difficult to know how much additional N fertilizer is needed to meet crop demands (Archontoulis et al., 2020; Grandy et al., 2022; McDaniel et al., 2020). It is common for most, if not all, fertilizer to be added prior to the growing season based on optimal growing conditions and yield potential, or fields are held near N saturation to ensure there is ample N for high grain yields (Grandy et al., 2022; McSwiney et al., 2010; Udvardi et al., 2021). This is problematic due to variable precipitation and irrigation availability in season, exacerbated by depleted freshwater aquifers, and shifting growing seasons making the frequency and severity of water limitations more challenging to predict (Deines et al., 2020; Derner et al., 2015; Rudnick, Irmak, Ray, et al., 2017).

The objective of this field study was to investigate maize response to different N and water availabilities and identify the underlying factors contributing to these responses. A better understanding of maize response can lead to improved N and water management thus improving yields and resource use efficiency. Specifically, we sought to: 1) quantify the agronomic optimum N availability for maize grain yield under different water availabilities across 3 growing seasons; 2) quantify total aboveground maize N uptake and determine if N availability and uptake are limiting yield when water is limited; and 3) examine maize resource use efficiency (water productivity, WP, and nitrogen use efficiency, NUE) in response to N and water availability.

2. Materials and methods

2.1. Site description

The experiment was conducted at the USDA-ARS Great Plains Research Center in Akron, Colorado (40°09 N, 103°09 W, 1383 m). The dominant soil type is a Weld silt loam (fine, smectitic, mesic Aridic Argiustoll). The field site is within a semiarid climate with average monthly temperatures ranging from 23 °C in the summer to −5 °C in the winter, and average annual precipitation of 420 mm occurring primarily from May – September (Schneekloth et al., 2020). In 2019 and 2020, the field was managed as a fully irrigated maize system with limited N applications to reduce residual soil N levels for the start of this study. Prior to this, the field had been managed using no-till practices since before 1990 for multiple dryland or irrigated crops. At the start of the 2021 growing season, baseline soil samples were collected from each block and sent to a commercial lab (Ward Laboratories, Inc; Kerney, NE). Samples were collected to a depth of 91 cm and separated into intervals of 0 – 15 cm, 15 – 61 cm, and 61 – 91 cm. The results showed that average soil pH ranged from 8 to 8.3 and soil organic matter ranged from 1.7 to 1.8%.

A field trial was established in 2021 with data collected during the 2021, 2022, and 2023 growing seasons. On May 6, 2021, a 104-day maturity maize hybrid (DKC 54–64) was planted at 79,000 seeds ha^{-1} and on May 10, 2022, the same maize hybrid was planted at 84,000 seeds ha^{-1} . On May 8, 2023, a 106-day maturity maize hybrid (DKC 56–65) was planted at 84,000 seeds ha^{-1} .

We utilized a split-plot experimental design with a total of 36 plots. Plots were 12 rows wide and approximately 37 m long with 0.76 m row spacing. There were 3 blocks with 2 water treatments randomly assigned within each block and 6 N availability treatments (sum of N fertilizer applied, N in irrigation water, and pre-planting soil inorganic N 0 – 90 cm depth) randomly assigned within each of those water treatments. Water treatments consisted of full water (100% ET), and limited water (target 70% ET), which is near average seasonal precipitation for the region (dryland conditions). During the 2021 and 2022 growing season the N treatments started at 22 kg N ha^{-1} and increased in increments of 50 – 51 kg ha^{-1} with the highest rate being 275 kg N ha^{-1} capturing low to excessive N additions. Soil residual inorganic N was much higher at the start of 2023 so N fertilizer application rates were reduced. When considering both residual N and applied N, N treatments were similar across the growing seasons. During the 2023 growing season the full water N rates were 22, 90, 106, 168, 212, and 247 kg N ha^{-1} . Residual N was higher in the limited water compared to full water especially in plots that had received higher N fertilizer rates the previous two growing seasons. Therefore, N rates were reduced to 22, 90, 106, 123, 123, and 202 kg N ha^{-1} for limited water plots during the 2023 season. At planting, liquid ammonium phosphate at a rate of 22 kg N ha^{-1} and 67 kg P ha^{-1} was added across all plots during the 2021 growing season. During the 2022 and 2023 growing seasons liquid ammonium phosphate at a rate of 22 kg N ha^{-1} and 44 kg P ha^{-1} was added at planting across all plots. The rest of the fertilizer N was side dressed at V6-V7 (6th-7th leaf vegetative stage) as urea ammonium nitrate (UAN, 32–0–0) banded to the soil surface approximately 5 cm from the row to create the different nitrogen treatments. Growth stages were documented with standard notation of stages for maize (Abendroth et al., 2011). Side dressed fertilizer was immediately followed by a watering event of 13 mm to incorporate it into the soil.

Soil inorganic N ($\text{NO}_3\text{-N}$ & $\text{NH}_4\text{-N}$) was measured before planting each season in the spring. In 2021, 3 soil samples were taken from each block with a tractor mounted hydraulic probe (Giddings, Windsor, CO) to a depth of 90 cm. Inorganic N amounts ranged from 24 kg ha^{-1} – 35 kg ha^{-1} . In 2022 and 2023, 2 samples were taken from each plot to a depth of 90 cm with the same hydraulic probe. Soil inorganic N ranged from 11 kg N ha^{-1} – 24 kg N ha^{-1} across the different treatments in 2022, and, from 29 to 124 kg N ha^{-1} in 2023. The plot treatments were

in the same location for each growing season, verified by GPS coordinates from the previous season. Spring residual inorganic N measurements was only used to adjust N application rates in 2023.

2.2. Evapotranspiration measurement and irrigation management

The soil water balance method (Eq. 1) was used for measuring actual crop evapotranspiration (ET_c act), hereafter referred to as ET, is described by (Schneekloth et al., 2020). Briefly, neutron probes (model 503DR, Concord, CA) were used to measure soil water content on a weekly basis (Bowman and King, 1965). Soil water measurements were done in 0.3 m increments to a depth of 1.8 m. Neutron probes were installed in the 6th row of each plot (the middle row) shortly after maize emergence and were removed after harvest.

A simplified water balance method was used to calculate ET (Bowman and King, 1965):

$$ET = PAV + IAV + \Delta MAV + (CAV + RAV + GAV) \quad (1)$$

where ET is actual crop evapotranspiration, PAV is total precipitation, IAV is net irrigation, CAV is total capillary rise from below rooting zone, ΔMAV is net change in soil water, RAV is net change in soil water due to surface flow, GAV is total percolation of water below the rooting depth. Drainage, capillary rise, and surface flow were assumed negligible. Using the water balance approach, we calculated the consumptive water use for the growing season.

Water was applied with a linear sprinkler system. The amount of water applied was based on growth stage and soil moisture content. The full water treatment avoided water stress, while the limited water treatment represented the long-term average annual precipitation amounts for dryland production in the region. This average corresponds to approximately 70% consumptive water use relative to the full water. The limited water treatment received more water during the reproductive stages, to reduce stress during this critical growth stage, compared to the vegetative stages, but still experienced some water stress during reproductive stages especially in 2022. The spring of 2021 was much wetter than the spring of 2022 allowing water applications to start later in 2021. The 2023 growing season had above average precipitation so the irrigation did not start until August, and the limited water treatment did not receive any supplemental irrigation. Weather data from CoAg-Met (Colorado Agricultural Meteorological Network, <https://coagmet.colostate.edu/>) was collected from an on-site weather station.

2.3. Grain yield, aboveground biomass, and plant nitrogen uptake

Aboveground maize nitrogen uptake was measured at silking (R1) and again at physiological maturity or black layer (R6). Maize N uptake was measured by collecting aboveground portions of 5 consecutive plants from each plot from one of the center 4 rows. No dry mass data is available for 2023 R1 leaves and stalks because it was not recorded. Dry stalk and leaf weights were estimated using the ratio of fresh stalks to dry leaf and ratio of dry leaf to fresh stalks from 2021, respectively, for all plots. Maize samples were separated by leaves, stalks, grain, and cobs, dried at 60 °C and weighed. Plant samples were then ground, homogenized, and a subsample was analyzed for total N concentration using a dry combustion elemental analysis (LECO Tru-SPEC, St. Joseph, MI, USA). Total N uptake was calculated by multiplying the oven dry biomass of each plant component by its corresponding N concentration and summing all the aboveground parts. Maize N uptake between R1 and R6 was calculated by taking the difference between N uptake from R6 samples and R1 samples. Maize grain yield and aboveground biomass were calculated at R6 using the same samples for N uptake. The grain was separated from the ears, weighed, measured for moisture content, and adjusted to 15.5% moisture. Maize N uptake, yield, and biomass was converted to a per hectare basis using the harvested area.

2.4. Belowground biomass

Maize root biomass was measured during the 2021 growing season after maize reached physiological maturity and had been harvested. Two soil cores per plot were taken with a 3.8 cm diameter tractor mounted hydraulic probe (Giddings, Windsor, CO): one in-row core, and one inter-row core. Soil cores were taken to a depth of 180 cm and were split into 7 depths (0 – 15, 15 – 30, 30 – 60, 60 – 90, 90 – 120, 120 – 150, and 150 – 180 cm). Samples were frozen until they could be processed. Samples were passed through a 2 mm sieve and roots were washed to remove soil and rocks from the sample. Root samples were then oven-dried at 60°C, weighed, and sub-replicated averaged for each plot on a ground area basis (e.g. horizontal surface area of the core).

2.5. Water productivity and nitrogen use efficiency

Grain yield and maize N uptake were used to calculate water productivity (WP) and two additional nitrogen use efficiency (NUE) metrics. Water Productivity was calculated using Eq. 2, where higher values indicate more grain yield per actual crop evapotranspiration (ET) used during the crop growing season (Comas et al., 2019).

$$\frac{\text{Grain Yield (kg grain ha}^{-1}\text{)}}{\text{Evapotranspiration (mm)}} \quad (2)$$

A partial N balance grain (PBN_{grain}) was calculated using Eq. 3. A PNB_{grain} specifically looks at N fertilizer applied compared to plant N uptake in the grain, which is exported from the system. Values > 1 indicate soil N mining, or more nitrogen being removed than added, and values < 1 indicate potential over application, or more N applied than what was accumulated by the plant (Congreves et al., 2021).

$$\frac{\text{Maize Nitrogen Uptake grain (kg N ha}^{-1}\text{)}}{\text{Nitrogen fertilizer rate (kg N ha}^{-1}\text{)}} \quad (3)$$

Maize N Utilization Efficiency (NUE) was calculated using Eq. 4, where higher values indicate more grain yield per unit nitrogen accumulated by the plant (Congreves et al., 2021).

$$\frac{\text{Maize Grain Yield (kg ha}^{-1}\text{)}}{\text{Maize Nitrogen Uptake (kg N ha}^{-1}\text{)}} \quad (4)$$

2.6. Statistical and optimum threshold analyses

A linear mixed effects model was used for all analyses to determine how the response variables (yield, aboveground biomass, belowground biomass, N uptake, WP, and NUE) were affected by water treatment, nitrogen availability (N fertilizer rate, soil residual N 0 – 90 cm, and N in applied water), year, and their interactions. Model selection was guided by Akaike Information Criterion (AIC) scores and visual inspection of model fit (i.e. graphed residuals and fitted values and quantile-quantile plots). The split plot design and repeated measures analyses were integrated into the mixed model structure through the inclusion of block and its interactions with water and N availability as random factors. Water availability and year were treated as categorical and N availability was treated as continuous factors. Response variables and N availability were checked for normality, equal variance, and homogeneity and were transformed to meet model assumptions. Grain yield, total aboveground biomass at R1 and R6, root biomass, total N uptake, N uptake from germination to R1, and N uptake from R1 – R6 had a quadratic relationship with N availability, which was modeled using a quadratic polynomial term. Maize WP and NUE were log transformed to meet model assumptions and N availability was modeled using a quadratic polynomial term. Maize PNB_{grain} was log transformed to meet model assumptions and N availability was modeled using a quadratic polynomial term. Maize belowground biomass was log transformed for both the entire soil profile and the top 0 – 60 cm to meet model assumptions.

An ANOVA with the "Kenward-Roger" degrees of freedom adjustment was used to assess the significance of the fixed effects and their interactions within the context of this mixed-effects model. To identify the agronomic optimum N level (AONR) and the corresponding yield at AONR, we used the fitted mixed-effects model to predict yield across a range of nitrogen availability values (sum of N fertilizer applied, N in irrigation water, and pre-planting soil inorganic N 0 – 90 cm depth). Based on model-generated response curves, the AONR was defined as the lowest N availability at which the predicted yield reached its maximum, and the corresponding predicted yield was recorded as the yield at AONR. The lowest N availability which where the response variable reached its maximum, was found for all response variables. Analyses were performed in R version 4.2.2 using the lme4, lmeTest, and emmeans packages (R Core Team, 2022).

3. Results

3.1. Precipitation, irrigation, and nitrogen amounts

The 10-yr average annual precipitation for the site is 420 mm with 313 mm occurring between May – October (Schneekloth et al., 2020). Total annual precipitation (January – December) for 2021 was near average with 421 mm of precipitation. However, 76% of the

precipitation was concentrated between March – May (Fig. 1a). Total precipitation in 2022 was 14% below average with 359 mm of precipitation. In 2022, precipitation was more evenly distributed across the growing season with 56% of precipitation occurring between May – August but this was still below average during this time frame. For 2023 precipitation was much higher than average with 577 mm of precipitation occurring. Precipitation was above average or near average from January – August (except for April) and 75% of the precipitation occurred between May – August.

For full water availability, 288, 470, and 115 mm of irrigation water was added during the 2021, 2022, and 2023 growing seasons respectively to meet crop ET demands. For the limited water availability treatment, 89, 199, and 0 mm of irrigation water was added during the 2021, 2022, and 2023 growing seasons respectively (Fig. 1a, S. Table 1). In the 2021, 2022, and 2023 growing seasons, total average ET in the limited water treatment was 76% (396 mm), 71% (556 mm), and 92% (582 mm), respectively, of the full water treatment, which was 518 mm, 782 mm, and 633 mm for each respective growing season.

The applied water had 8 ppm NO_3^- , which was equivalent to adding 2 kg N ha^{-1} per 25 mm of water applied. This added 23 and 7 kg N ha^{-1} for the full and limited water, respectively, in 2021. In 2022, an additional 38 and 16 kg N ha^{-1} was added from the applied water for the full and limited water, respectively. In 2023, an additional 9 kg N ha^{-1} was

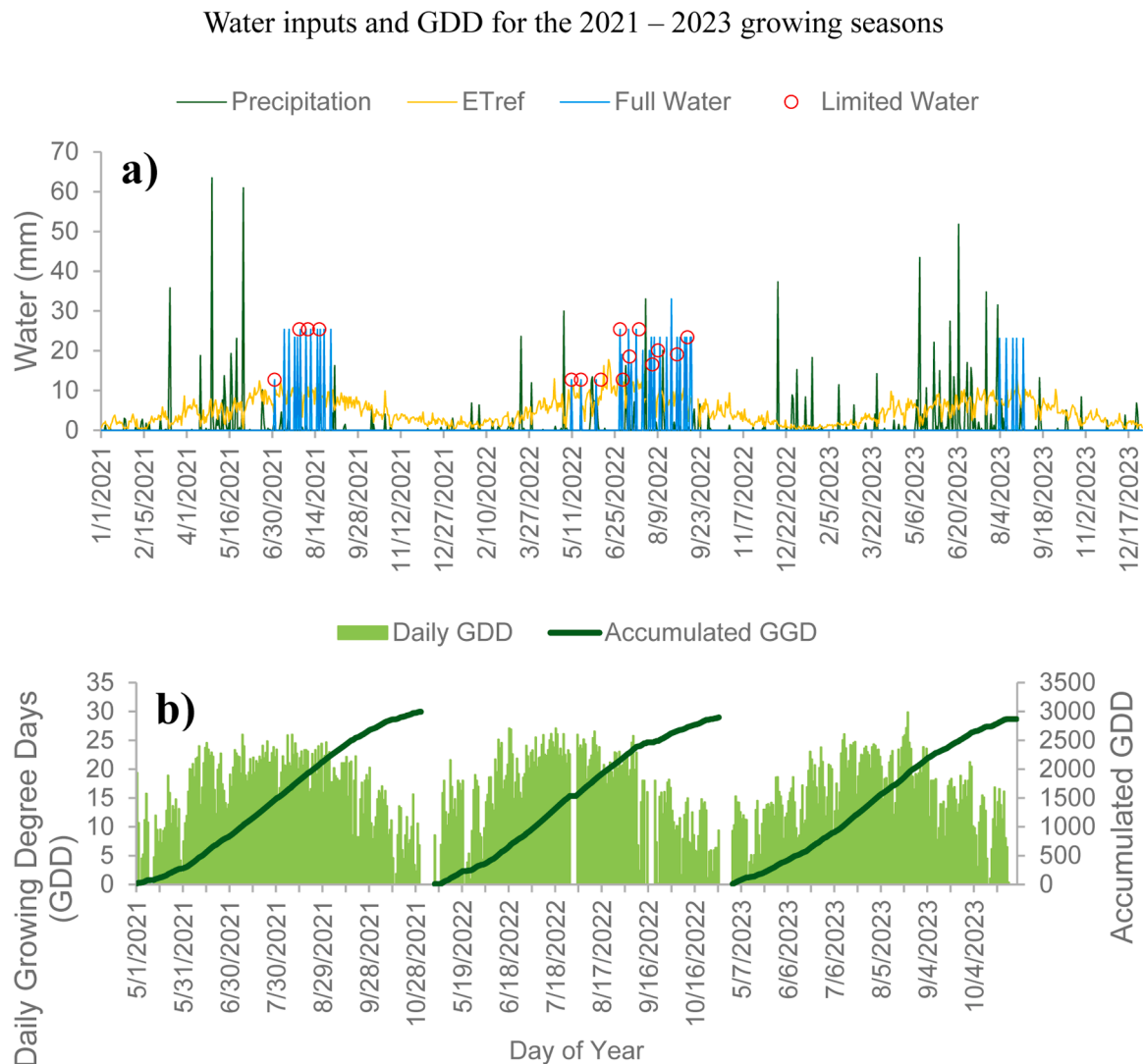


Fig. 1. a) Daily precipitation, full and limited water irrigation additions, and reference evapotranspiration (ET) for the three growing seasons of the experiment. b) Daily and accumulated growing degree days (GDD) during the three growing seasons of the experiment.

added to the full water treatment from water applied, with no additional N (0 kg N ha^{-1}) added to limited water due to lack of irrigation that year. The 2021 growing season total N availability (the sum of $\text{NO}_3\text{-N}$ and $\text{NH}_4\text{-N}$ from N fertilizer applied, soil residual N, and N from irrigation water) ranged from 74 to 327 kg N ha^{-1} for full water and $58 - 311 \text{ kg N ha}^{-1}$ for limited water. In 2022, total N availability ranged from 79 to 326 kg N ha^{-1} and $50 - 312 \text{ kg N ha}^{-1}$ for full and limited water, respectively. In 2023, total N availability ranged from 67 to 310 kg N ha^{-1} and $59 - 302 \text{ kg N ha}^{-1}$ for full and limited water, respectively (Table 1).

3.2. Grain yield and aboveground biomass

Maize grain yields increased curvilinearly with N availability but were dependent on water availability resulting in an $\text{N} \times \text{water}$ interaction (Fig. 2, $P < 0.01$). Maximum grain yield, and the AONR to achieve it (N availability at which grain yield is maximized) varied from year to year, especially for limited water resulting in an $\text{N} \times \text{year}$ interaction ($P = 0.01$). With full water availability during the 2021 and 2022 growing seasons, grain yield increased with greater nitrogen (N) availability and did not reach a clear plateau. The AONR was 327 kg N ha^{-1} in 2021 and 326 kg N ha^{-1} in 2022 which was the highest N availability used. In 2023, the AONR for grain yield under full water was slightly lower, but close to the maximum N availability, at 286 kg N ha^{-1} . Across all three growing seasons with limited water, grain yield increased curvilinearly with N availability up to the AONR, after which it declined. The AONRs under limited water were 223, 199, and 219 kg N ha^{-1} for 2021, 2022, and 2023, respectively, which is near the 4th N treatment (Table 1).

When water was limited, N availability could be reduced by an average of 31% (32%, 39%, and 23% for the 2021, 2022, and 2023 growing seasons respectively) relative to full water to achieve maximum grain yields. However, with limited water maximum grain yields declined by 36%, 59%, and 30% relative to full water for the 2021,

2022, and 2023 growing season respectively.

A $\text{water} \times \text{year}$ interaction was observed ($P < 0.01$) where maximum grain yields were nearly identical with full water availability across the three growing seasons. However, with limited water availability, maximum grain yields in 2023 were 73% and 13% greater than 2022 and 2021, respectively. In 2021, 2022, and 2023, maximum grain yields with limited water were achieved with 74% (404 mm vs. 545 mm), 74% (566 mm vs. 770 mm), and 93% (595 mm vs. 642 mm) of the full water ET, respectively.

Aboveground biomass at R6 followed a similar trend as grain yield where biomass increased curvilinearly with N rate and was dependent on water availability as well as year, resulting in an $\text{N} \times \text{water}$ and $\text{N} \times \text{Year}$ interaction (S. Fig. 1, $P = < 0.01$ & $P < 0.01$, respectively). The AONR (N availability that maximized aboveground biomass) was achieved near the highest N availability (303, 290, and 260 kg N ha^{-1} for 2021, 2022, and 2023 respectively) for full water, but near the 4th N availability for limited water (232, 216, and 207 kg N ha^{-1} for the 2021, 2022, and 2023 growing season respectively). With limited water, aboveground biomass decreased as N availability increased beyond the AONR. Aboveground biomass at R1 increased curvilinearly with N availability and the effect of water availability was dependent on growing season resulting in a $\text{water} \times \text{Year}$ interaction (S. Fig. 2, $P < 0.01$ & $P = 0.01$ respectively). For both the 2021 and 2022 growing seasons, the AONR was 227 kg N ha^{-1} when water was limited, and biomass decreased as N availability increased beyond the AONR. Full water increased maximum aboveground biomass at R1 by 29% and 7% compared to limited water for the 2021 and 2022 growing season respectively.

3.3. Belowground biomass

For the entire soil profile (0 – 180 cm) there was no water nor N effect (S. Fig. 3, $P = 0.38$ & 0.89, respectively). For the entire profile, there was a depth effect where average root biomass was greatest in the

Table 1

Total nitrogen (N) availability (the sum of $\text{NO}_3\text{-N}$ and $\text{NH}_4\text{-N}$ from N fertilizer applied, soil residual N, and N from irrigation water, kg N ha^{-1}) and the sources for all treatments and growing seasons. Soil residual N was measured in the spring prior to planting to a depth of 90 cm.

Nitrogen treatment	Full water 2021				Limited water 2021			
	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability
	(kg N ha ⁻¹)				(kg N ha ⁻¹)			
1	29	22	23	74	29	22	7	58
2	29	73	23	125	29	73	7	109
3	29	123	23	175	29	123	7	159
4	29	174	23	226	29	174	7	210
5	29	224	23	276	29	224	7	260
6	29	275	23	327	29	275	7	311
Nitrogen treatment	Full water 2022				Limited water 2022			
	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability
	(kg N ha ⁻¹)				(kg N ha ⁻¹)			
1	19	22	38	79	12	22	16	50
2	17	73	38	128	12	73	16	101
3	15	123	38	176	14	123	16	153
4	18	174	38	230	16	174	16	206
5	14	224	38	276	24	224	16	264
6	13	275	38	326	21	275	16	312
Nitrogen treatment	Full water 2023				Limited water 2023			
	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability	Soil residual nitrogen	Nitrogen fertilizer rate	Irrigation water nitrogen	Total Nitrogen availability
	(kg N ha ⁻¹)				(kg N ha ⁻¹)			
1	36	22	9	67	37	22	0	59
2	38	90	9	137	35	90	0	125
3	38	106	9	153	43	106	0	149
4	30	168	9	207	68	123	0	191
5	34	212	9	255	124	123	0	247
6	54	247	9	310	100	202	0	302

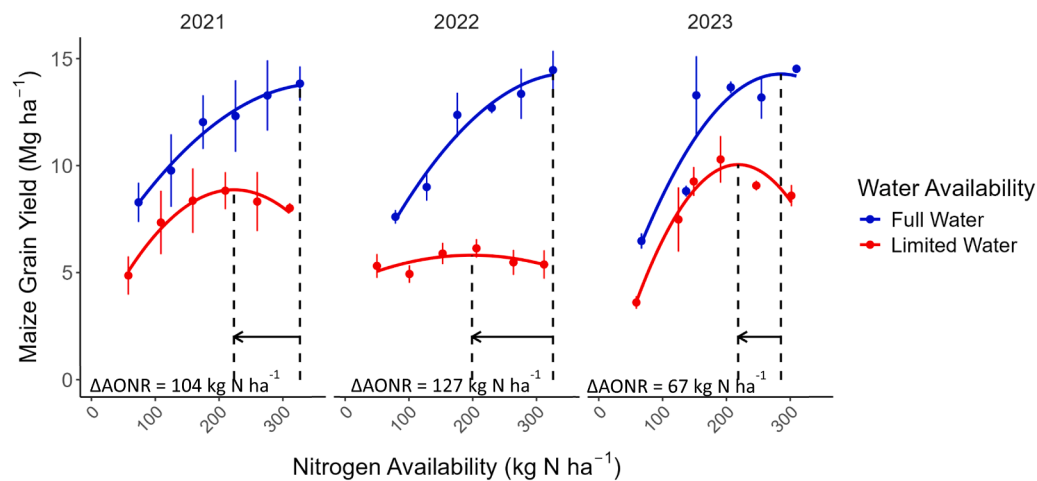


Fig. 2. Maize grain yield by growing season, and water and nitrogen availability. Maize grain yield was adjusted to 15.5% moisture “R6” corresponds to physiological maturity which is when samples were collected. Dotted lines indicate agronomic optimum N rate (AONR) for each water treatment. The AONR is defined as the lowest N availability where grain yield reached its maximum. Nitrogen availability is defined in Table 1. Error bars depict $\pm$ SE.

top 0 – 15 cm and decreased with soil depth.

When looking at the top 0 – 60 cm depths, which on average had 84% of the total root biomass for the entire profile, there was a 3-way interaction between $N \times \text{water} \times \text{depth}$ (Fig. 3, $P = 0.04$). With limited water root biomass was maximized with N availabilities of 144, 311, and 137 kg N ha⁻¹ for the 0 – 15, 15 – 30, and 30 – 60 cm depths respectively. With full water root biomass was maximized with 241, 203, and 327 kg N ha⁻¹ of N availability for the 0 – 15, 15 – 30, and 30 – 60 cm depths. Maximum root biomass was 68%, 19%, and 24% greater with limited water compared to full water for the 0 – 15 cm, 15 – 30 cm, and 30 – 60 cm depths respectively.

3.4. Aboveground maize nitrogen uptake

Maize N uptake for the entire season (from germination – physiological maturity at R6) increased curvilinearly with N availability, and was dependent on water availability resulting in an $N \times \text{water}$ interaction (Fig. 4, $P = 0.04$). Maximum N uptake was achieved at the highest

N availability at the highest N availability expect for limited water during the 2023 growing season which maximized N uptake at 282 kg N ha⁻¹. Interactions between N availability and year as well as year and water availability were observed ($P = < 0.01$ & < 0.01). Maximum N uptake was 45%, 24%, and 71% greater with full water relative to limited water for the 2021, 2022, and 2023 growing seasons respectively. Total aboveground maize N uptake exhibited a strong, positive relationship with grain yield and aboveground biomass, especially for full water and when focusing on the lower N availabilities (N treatment 1 – 4; S. Table 2 & 3). This relationship became weaker or non-existent at the higher N availabilities (N treatment 4 – 6).

Maize N uptake was also examined from germination to R1 and from R1 to R6 in 2021 and 2022. Maize N uptake from germination to R1 exhibited an $N \times \text{water}$ interaction where N availability increased N uptake especially with full water (Fig. 5a & c, $P = 0.06$). This $N \times \text{water}$ interaction was more pronounced during the 2021 growing season. The maximum N uptake was achieved at the highest N availability for both full and limited water, however full water increased N uptake by 41%

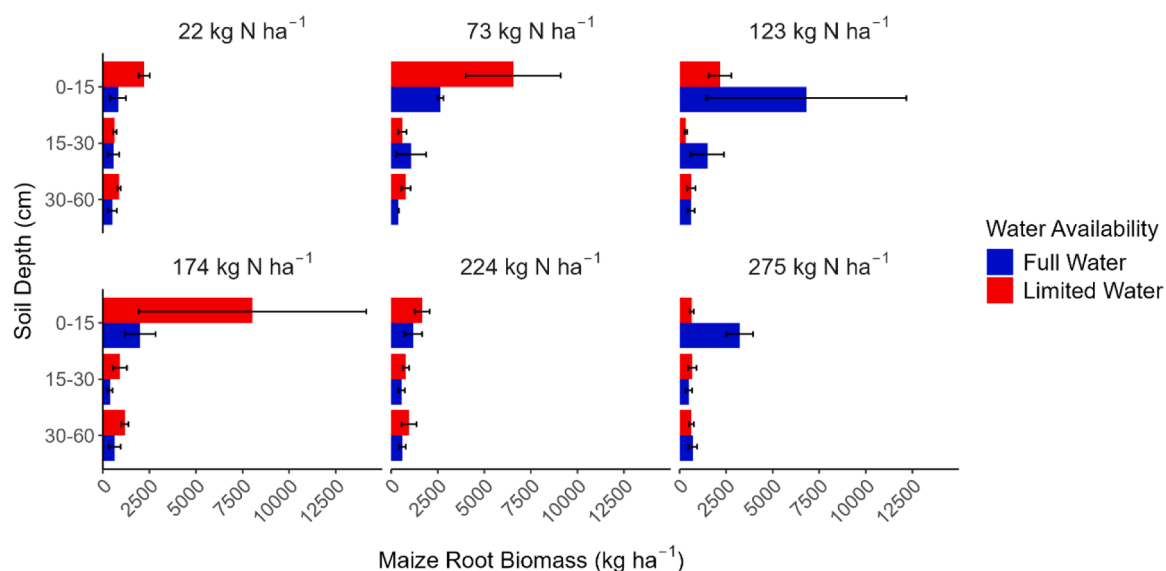


Fig. 3. Maize root biomass for the 2021 growing season by water availability, soil depth increment, and nitrogen (N) fertilizer rate. The N fertilizer rates ranged from 22 to 275 kg N ha⁻¹ and the total N availability (sum of N fertilizer rate, pre-plant soil N, and N in irrigation water applied) ranged from 58 to 327 kg N ha⁻¹ and are defined in Table 1. Root samples were collected in the fall after maize reached physiological maturity and was harvested. Total N availability varied between the two water availabilities. Therefore, N fertilizer rate was used in the heading to be consistent across the treatments. Error bars depict $\pm$ SE.

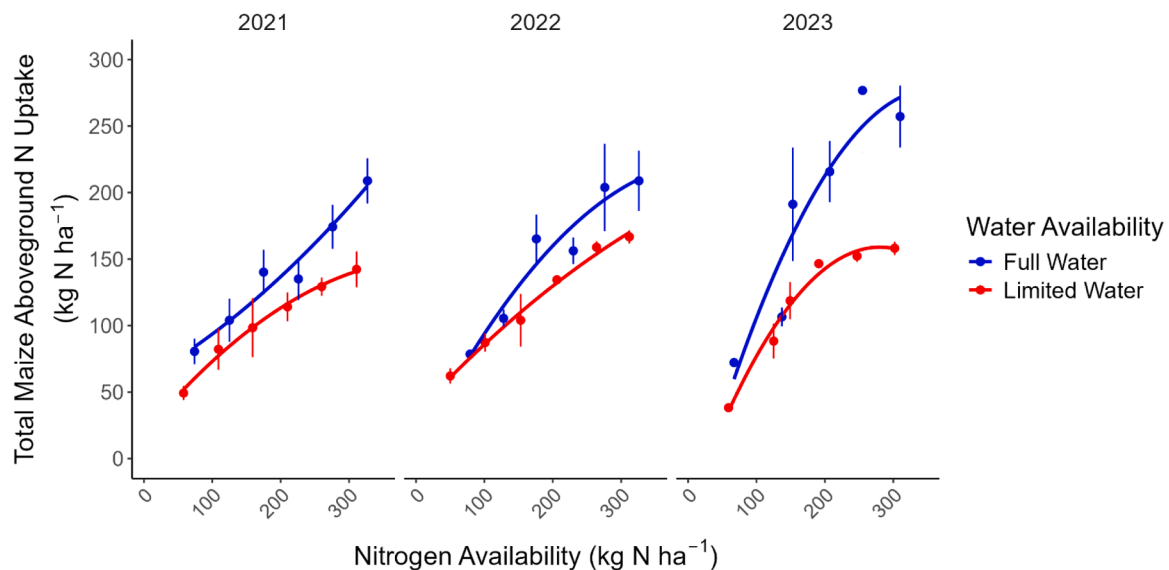


Fig. 4. Total aboveground maize nitrogen (N) uptake over entire growing season (germination – physiological maturity) by growing season, water and nitrogen availability. Error bars depict $\pm$ SE.

Aboveground maize nitrogen uptake by growing season, growth stages, water, and nitrogen availability

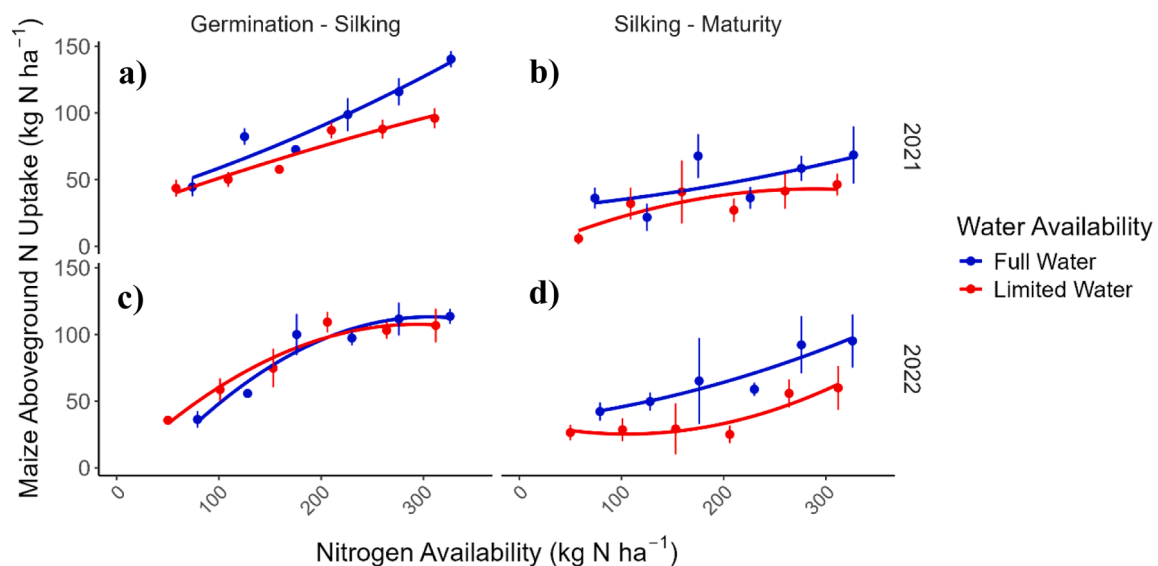


Fig. 5. a) Aboveground maize nitrogen (N) uptake from germination – silking (R1) by water and nitrogen availability for the 2021 growing season. b) Aboveground maize nitrogen (N) uptake from R1 – physiological maturity (R6) by water and nitrogen availability for the 2021 growing season. c) Aboveground maize nitrogen (N) uptake from germination – R1 by water and nitrogen availability for the 2022 growing season. d) Aboveground maize nitrogen (N) uptake from R1 – R6 by water and nitrogen availability for the 2022 growing season. Error bars depict $\pm$ SE.

compared to limited water in the 2021 season. During the 2022 season, maximum N uptake was achieved at 307 and 294 kg N ha⁻¹ for full and limited water respectively, and full water increased maximum N uptake by 5% thus there was a water $\times$ year effect ($P = 0.01$). The N rate that maximized N uptake varied by year leading to an N $\times$ year interaction ($P = 0.04$). From R1 – R6, maize N uptake increased curvilinearly with N availability and was higher with full water availability (Fig. 5b & d, $P < 0.01$ & 0.09). Maize N uptake between R1 and R6 was the greatest at or very near the highest N availability for both full and limited during both growing seasons. No N $\times$ water interaction was observed during these later growth stages ($P = 0.87$). Maize N uptake between R1 – R6

was 30% greater in 2022 than 2021 ($P = 0.02$).

3.5. Water productivity

Maize WP exhibited a three-way interaction between N, water availability and growing season resulting in an N $\times$ water $\times$ year interaction (Fig. 6, $P = 0.04$). With full water availability, maximum WP was achieved with 323, 326, and 271 kg N ha⁻¹ for the 2021, 2022, and 2023 growing seasons respectively. When water was limited, maize WP was maximized with 214, 198, and 223 kg N ha⁻¹ for the 2021, 2022, and 2023 growing seasons. Maize WP decreased as N availability

Maize water productivity by growing season, water, and nitrogen availability

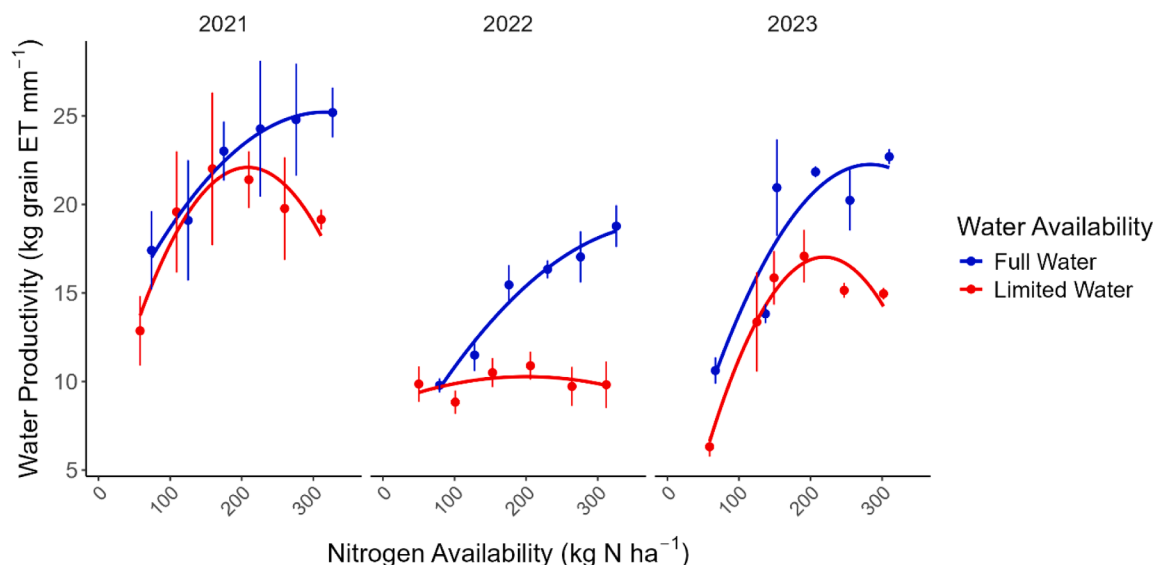


Fig. 6. Maize water productivity by growing season, water availability, and nitrogen availability. Water productivity was calculated as grain yield (kg ha^{-1}) divided by actual crop ET (mm) over the entire season. Error bars depict $\pm$ SE.

increased beyond the AONR. Across the three growing seasons, the maximum WP with full water was greatest in the 2021 growing season and was 37% and 12% greater than the 2022 and 2023 growing season respectively. With limited water, the maximum WP was 116% and 24% greater in the 2021 growing season relative to the 2022 and 2023 growing seasons. Full water availability resulted in 14%, 80%, and 27% greater maximum WP relative to limited water for the 2021, 2022, and 2023 growing seasons respectively.

3.6. Nitrogen use efficiency metrics

Maize $\text{PNB}_{\text{grain}}$ (partial N balance grain; ratio of N in grain to N fertilizer applied) was greater with full water availability, and the effect of N rate was dependent on growing season, resulting in an $\text{N} \times \text{year}$ interaction (Fig. 7a, $P = 0.02$ & $P < 0.01$). On average, maize $\text{PNB}_{\text{grain}}$ was 49%, 56%, and 25% greater in full water compared to limited water for the 2021, 2022, and 2023 growing season respectively. The 2023 growing season had greater $\text{PNB}_{\text{grain}}$ values compared to the 2022 and 2021 growing seasons, especially at higher N fertilizer rates. The 2023 growing season had elevated soil residual N concentrations before planting leading to reduce N fertilizer rates (Table 1). Maize $\text{PNB}_{\text{grain}}$ was always maximized at the lowest N availability and showed soil N mining was occurring, while higher N availability led to lower $\text{PNB}_{\text{grain}}$ values and showed more N was added than what plants maize acquired in the grain. Maize $\text{PNB}_{\text{grain}}$ was often balanced ($\text{PNB}_{\text{grain}} = 1$) between the two lowest N rates for both water treatments. This was true except for the 2022 and 2023 growing seasons which were balanced between the 2nd and 3rd N fertilizer rate and 3rd and 4th N fertilizer rates for the 2022 and 2023 growing season respectively.

Maize NUtE (nitrogen utilization efficiency; ratio of grain yield to N uptake) decreased with increased N availability and exhibited a water $\times$ year interaction (Fig. 7b, $P < 0.01$ & $P < 0.01$). Maize NUtE was always maximized at the lowest N availability, decreasing sharply from the lowest to second lowest N availability and had progressively smaller declines until it reached its lowest at the highest N availability. However, maximum NUtE values varied by growing season. For the 2021 growing season, NUtE was 4% greater with full water relative to limited water. During the 2022 growing NUtE was 18% greater with full water, however during the 2023 season limited water increased the maximum

NUtE by 3% relative to full water (Fig. 7b).

4. Discussion

Improving grain yields with limited water is becoming increasingly important as water limitations become more frequent (Deines et al., 2020; Derner et al., 2015; Nilahyane et al., 2023). Past research suggests that water limitation can reduce soil N availability and supply, and plant N uptake, thereby limiting plant growth, and that increasing N inputs may improve crop performance under drought (Ashraf et al., 2016; Maywald et al., 2022; Zulfiqar et al., 2017). The results of the current multi-year study do not support this assumption that additional N availability can help ameliorate the negative effects of water stress on maize. Instead, high N availability reduced maize grain yield and aboveground biomass under limited water, in alignment with some previous studies in maize and winter wheat (Al-Kaisi and Yin, 2003; Flynn et al., 2023; Mon et al., 2016). The current study consistently demonstrated across three growing seasons that although maize N uptake increased with N availability, both pre- and post-silking, this did not translate into yield gains under water limitation. This decoupling of N uptake from yield formation reveals that N acquisition was not the primary yield-limiting factor.

Given the economic and environmental effects of water and N use, there is growing interest in improving water and N management at local and regional levels (Ferin et al., 2026; Rudnick et al., 2020; Stockton and Sand, 2025) as well as global goals sponsored by the United Nations (Chen et al., 2026). To achieve maximum yields and resource use efficiency, N and water inputs need be adjusted according to yield potential. With limited water, the amount N should be reduced by, and the reduction in yield will depend on many factors, especially soil N availability and water stress timing and severity.

4.1. Maize agronomic optimum nitrogen rate

Across the 3 growing seasons, the AONR for grain yield was relatively constant for both limited and full water treatments. For limited water, the AONR for grain yield ranged between 199 – 223 kg N ha^{-1} , while the AONR with full water was 286 – 327 kg N ha^{-1} , representing an average 31% reduction. However, the lack of a yield plateau in full

Maize nitrogen use efficiency metrics by growing season, water, and nitrogen availability

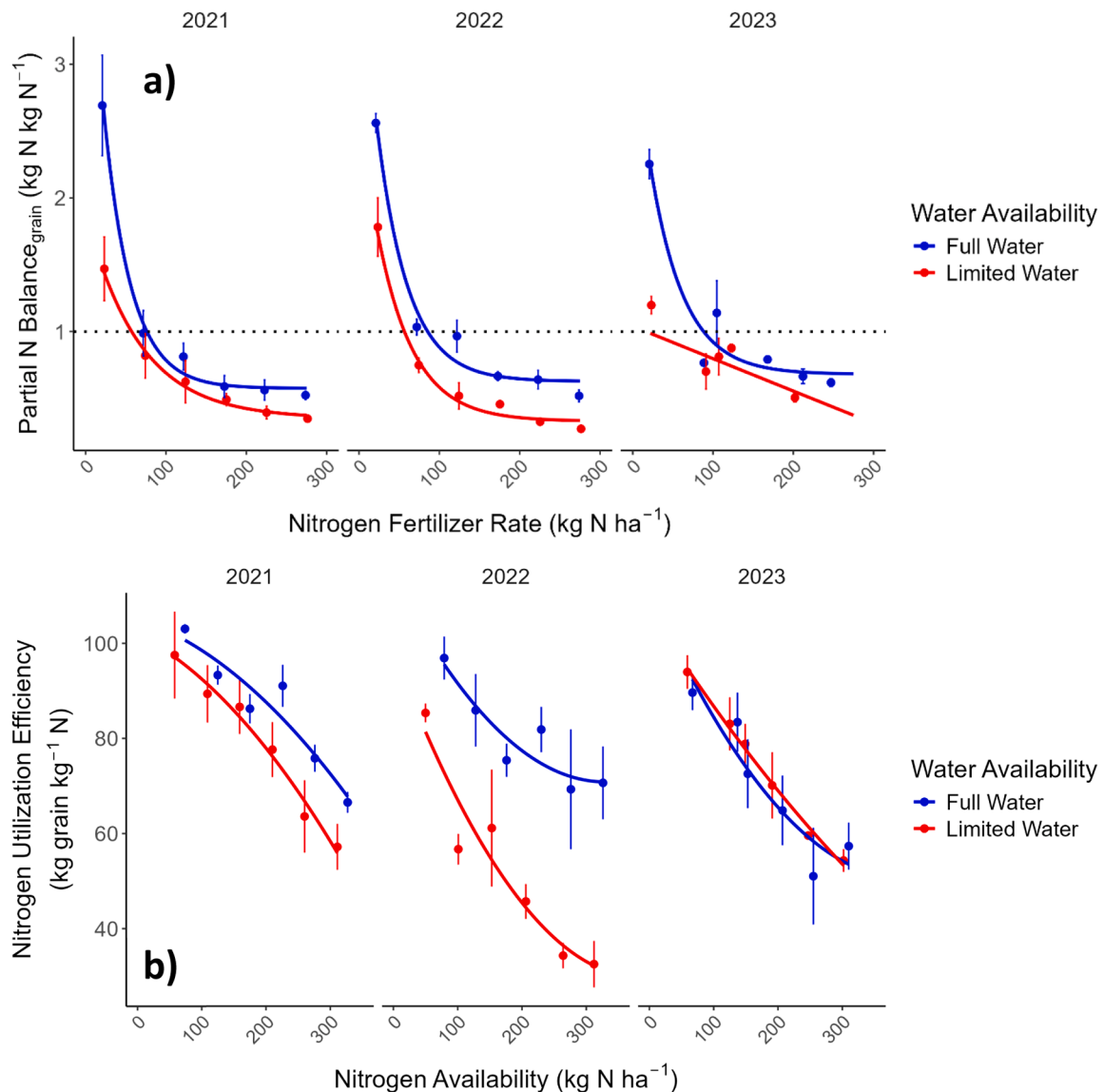


Fig. 7. a) Maize nitrogen use efficiency (NUE) partial nitrogen (N) budget grain (PNB_{grain}) by growing season, water availability, and N fertilizer rate. Maize NUE PNB_{grain} was calculated by dividing maize N uptake in the grain only (kg N ha⁻¹) at physiological maturity (R6) by N fertilizer applied (kg N ha⁻¹). Values above the dotted black line show more N is accumulated in the grain and removed at harvest relative to N applied as fertilizer. Values below 1 show that more N fertilizer was added relative to N removed. Values equal to 1 show a balanced N fertilizer budget (amount of N removed at harvest is equal to N added as fertilizer). Maize NUE PNB_{grain} uses N fertilizer instead of N availability (sum of N fertilizer, pre-plant residual N, and N in irrigation water) on the x-axis because it is specifically a fertilizer-based index for NUE. Nitrogen fertilizer rates are defined in Table 1 and 2023 had lower N rates due to higher residual soil N before planting. b) Maize nitrogen utilization efficiency (NUE) by growing season, water availability, and nitrogen availability. Maize NUE is calculated as grain yield (kg ha⁻¹) divided by total aboveground maize N uptake (kg N ha⁻¹) at R6. Higher values indicate greater yield per unit N uptake. Error bars depict $\pm$ SE.

water treatments suggests that the AONR may be higher than the rates included in this study. In 2023 soil residual N in the top 0–90 cm ranged from 30 to 124 kg N ha⁻¹ and only 13–29 kg N ha⁻¹ for the 2022 and 2021 growing seasons. This led to N fertilizer rates being reduced in the 2023 season (Table 1). Even though N fertilizer rates were reduced in 2023, yields were very similar for full water across the years and were greater in 2023 with limited water relative to 2021 and 2022. Additionally, in the 2023 season, there was still a yield decline at the highest N rates for limited water even though the limited treatment received an estimated 92% ET due to increased precipitation inputs.

The AONR for grain yield for limited water was, on average, 31% lower compared to full water. Across all three growing seasons, a

consistent threshold N rate emerged under limited water, beyond which additional N did not improve yield but reduced it. However, the magnitude of yield declines varied by growing season (36%, 59%, and 30% lower for 2021, 2022, and 2023, respectively). These differences in yield are likely a reflection of the different growing seasons water conditions relative to the full water treatment (25%, 27%, and 7% less ET in 2021, 2022, and 2023 respectively). For example, the highest yields with limited water occurred during the 2023 growing season. This season had above average precipitation throughout most of the growing season, thus no supplemental irrigation was added to the limited water treatment and the full water treatment only received weekly irrigation during the month of August. This additional water led to large yield

gains as stress was reduced during reproductive and grain fill stages. In contrast, the 2022 growing season had the greatest amount of irrigation applied, but also had the highest evaporative demand and lowest yield. Irrigation was added during early reproductive stages and throughout grain fill, however water demands were never fully met for the limited water treatments. While early reproductive stages are most susceptible to water stress, and strategic deficits can maintain high yields while saving water (Comas et al., 2019; Zou et al., 2021), water stress at all growth stages can decrease yields especially if severe water deficits persist through lack of precipitation or irrigation. Water stress reduces potential kernel number, photosynthates availability, anthesis and pollination, and translocation of photosynthates to new organs including kernels during grain fill ultimately decreasing yields (Aslam et al., 2015; Deribe, 2025). Collectively, the results show that both severity and timing of deficits govern yield response under water limited conditions.

4.2. Maize nitrogen uptake and the negative effects of high N

The hormetic effect of N, initially beneficial but eventually becoming detrimental (Schirrmacher, 2021), with limited water was observed in all 3 growing seasons in the current study. While water limitations reduce N uptake and availability of soil N, plant N uptake can persist through diffusion of N through the root zone, high soil N concentration from high N inputs, and recovery of plant N uptake after soil re-wetting (Buljovic and Engels, 2001; Gonzalez-Dugo et al., 2010; Wang et al., 2025). Maize N uptake both pre- and post-silking often has a positive relationship with yield as it increases leaf chlorophyll content, leaf longevity, photo-assimilate production, kernel weight and grain yield (Ciampitti et al., 2013; Ciampitti and Vyn, 2011; Hammad et al., 2017). High N uptake early in the season has also been shown to buffer against limited N uptake post-silking by meeting plant N demands through increased N remobilization (Nasielski et al., 2019). In the current study, maize N uptake increased with N availability both pre- and post-silking and was greater with full water. With full water, maize grain yield increased with N uptake. However, with limited water grain yield was maximized at a more modest N availability, and yields were depressed at the higher N availabilities despite higher N uptake.

One key mechanism contributing to yield declines with high N availabilities and limited water appears to be altered root growth. In the upper profile (0 – 60 cm), where a majority of root biomass and water extraction occur, excess N reduced root biomass under limited water, particularly in the surface (0 – 15 cm) and 30 – 60 cm depths. Previous studies have shown that excess N can reduce root biomass and alter root architecture towards a greater proportion of fine roots relative to coarse roots (Chen et al., 2015; Drobniitch et al., 2026; Ordóñez et al., 2021). Reducing root investment likely limited water acquisition from both shallow wetting fronts (precipitation/irrigation) and deeper stored soil water (Bauder et al., 2020; Comas et al., 2013; Jiang and Whalen, 2025), intensifying water stress during critical reproductive and grain-filling stages.

A second potential mechanism for decreased maize grain yield at the higher N availabilities with limited water is due to nitrogen's effect on stomata regulation (Cramer et al., 2009; Ding et al., 2018), which has been identified as an important trait in N × water interactions (Drobniitch et al., 2024). Elevated N concentrations can decrease stomatal conductance, thus reducing photosynthesis, carbohydrate production, and ultimately yield (Flynn et al., 2023; Ru et al., 2022; Wang et al., 2020). Reducing stomatal conductance can also increase heat stress as transpiration is decreased and evaporative cooling declines (Wahid et al., 2007).

A third possible mechanism that may have led to yield declines with high N availability and limited water is the disruption of source–sink relationships and the energetic cost of N assimilation. Water deficits disrupt source–sink relations by decreasing water movement throughout plants, reducing cell turgor pressure, and increasing phloem viscosity,

thereby reducing sugar export to sink organs and limiting sink strength (Rodrigues et al., 2019; Tegeder and Masclaux-Daubresse, 2018). As a result, sugars accumulate in source organs, which can downregulate photosynthetic activity further constraining the supply of photosynthates (Muller et al., 2011; Rodrigues et al., 2019). At the same time, assimilation of inorganic N into amino acids is an energy-intensive process requiring substantial ATP, reducing power, and carbon skeletons produced by photosynthesis (Hu et al., 2025; Xu et al., 2012), and water stress further constrains this process by reducing nitrate reductase activity, leading to reduced N assimilation and remobilization (Dinh et al., 2017; Guo et al., 2026; Xing et al., 2019). Together these results may indicate that under water limitation and high N availability and uptake, photosynthate supply and translocation are constrained while energetic demand for N assimilation remains high. This leads to a decoupling of N uptake and grain yield production as N uptake exceeds the plant's ability to efficiently assimilate and utilize it for grain production.

Ultimately, these results indicate that, although maize continued to accumulate N with increasing N availability under water limited conditions, grain yield was maximized at lower N rates. This divergence demonstrates a decoupling between N uptake and yield formation, suggesting that N supply and uptake was not the primary limiting factor for yield at higher N rates. This highlights that under water limitation, increasing N availability alone is insufficient to improve productivity as carbon assimilation and physiological processes may become the dominant constraints.

4.3. Water productivity

The effect of N availability on water productivity (WP) in water-limited systems is variable in the literature. Some studies report increases in WP with N (Mansouri-Far et al., 2010; Ogola et al., 2002; Pandey, Maranville, and Chetima, 2000), while others report decreases (Wang et al., 2017), with the responses often differing by growing season (Al-Kaisi and Yin, 2003; Li et al., 2025; Rudnick and Irmak, 2013). In the current study, moderate N supply improved WP, whereas excess N consistently reduced WP across all growing seasons for limited water.

Because WP is determined by the ratio of grain yield to ET, reductions in yield will decrease WP unless accompanied by proportional reductions in ET (Hatfield and Dold, 2019; Ullah et al., 2019). In the current study, high N reduced grain yield and above ground biomass when water was limited without proportional decreases in ET leading to low WP. Reduced biomass could have increased soil evaporation, as sparse canopies increase evaporative losses from the soil (Lenka et al., 2009; Ogola et al., 2002; Rudnick, Irmak, Djaman, et al., 2017). Consequently, water loss via evaporation was not offset by gains in biomass or yield, leading to low WP. Additionally, the 2022 growing season had the highest ET and irrigation inputs without the highest yields, thus WP was reduced. Improving WP is critical as water stress and diminished water resources become more common. Proper N fertilizer rates paired with management practices that reduce soil evaporation, such as synchronizing water inputs with crop water needs and retaining crop residue (Quemada and Gabriel, 2016; Schneekloth et al., 2020), and selecting desirable plant traits for water stress tolerance may become more common in order to improve WP (Comas et al., 2025; Danalatos et al., 2026).

4.4. Nitrogen use efficiency metrics

Maize NUE can be broken into two parts: uptake and utilization (Asibi et al., 2019). For N uptake, maize partial N balance grain (PNB_{grain}) is one metric that can be used. Maize PNB_{grain} is a fertilizer-based NUE index and is calculated as the ratio of N uptake by the grain, that is ultimately removed from the system, relative to the N fertilizer rate applied. Values greater than 1 indicate soil mining as more N is removed than applied, while low values indicate excessive N may

have been applied (Congreves et al., 2021). Ideally, high grain yield would be achieved while having a balanced system. In the present study, PNB_{grain} values were greater during the 2023 season due to N fertilizer rates being reduced. Improving N uptake can increase PNB_{grain} and other NUE metrics and can help mitigate N losses to the environment (Asibi et al., 2019; Ojeda-Rivera et al., 2025). Higher N fertilizer rates inevitably lead to lower NUE as yield increase per unit N diminishes, and N losses increase at higher rates (Davies et al., 2020; Puntel et al., 2016; Venterea et al., 2012). Management practices aimed at synchrony between plant N demand and soil N mineralization as well as N fertilizer additions, could reduce N fertilizer inputs while still meeting crop N demands thus increasing NUE (Loecke et al., 2012; Quemada and Gabriel, 2016; Udvardi et al., 2021). The inherently high mobility of mineral N in soil will always make synchrony between crop N uptake and soil N mineralization challenging and has motivated research focused on managing N availability from more stable soil N pools (Breza et al., 2023; Franzluebbers, 2016; Grandy et al., 2022).

Maize nitrogen utilization efficiency (NUE) is a measurement of how efficiently plants convert the N they acquired into yield (Asibi et al., 2019; Congreves et al., 2021). Full water increases NUE relative to limited water due to greater assimilation and remobilization of N, availability of photosynthates, ATP, and NADPH, and normal hormone and enzyme activity (Guo et al., 2026; Nasielski et al., 2019; Sun et al., 2025). Given the disconnect between N uptake and yield in the present study, improving utilization of N, rather than N uptake alone may be important for increasing NUE in water limited environments.

4.5. Future directions

The observed decoupling of N uptake from yield under combined high N and water limitation highlights the need for deeper mechanistic understanding of root hydraulics, stomatal behavior, and C-N interactions in maize (Comas et al., 2013; Drobniitch et al., 2024). carbon costs of N assimilation under varying water regimes, explore genotypic variation in root plasticity and stomatal sensitivity, and test in-season N management that adjusts rates according to real-time water availability (Amori et al., 2026; Guo et al., 2026; Mandić et al., 2025). Isotopically labeled N approaches could further clarify N sources, losses, and remobilization (Nasielski et al., 2019; Poffenbarger et al., 2018). Extending these findings to other cropping systems integrating them with deficit-irrigation strategies will support adaptive nutrient management as water scarcity intensifies (Almeida et al., 2026; Gomez et al., 2025; Jiao et al., 2024).

In the current study, the AONR for grain yield and aboveground biomass with full water was at or near the highest N availability suggesting that the true optimum may not have been reached. Future field studies should evaluate higher N availabilities to better define the upper limits of crop N demand. However, AONR is often higher than the economic optimum N rate and environmental N rate (Baum et al., 2025; Poffenbarger et al., 2017). Therefore, maximizing yield may not always be the most desirable outcome.

5. Conclusion

Matching N supply to water-limited yield potential through moderate N availability maximizes maize productivity, water productivity, and nitrogen-use efficiency while avoiding the ecophysiological penalties associated with excess N. Accounting for residual soil N, in-season soil N supply, real time plant N status, and accurate water availability forecasting may become increasingly important for ensuring there is adequate N to maximize productivity and efficiency while minimizing environmental and economic losses.

CRedit authorship contribution statement

Joel Schneekloth: Writing – review & editing, Resources, Project

administration, Methodology, Investigation, Conceptualization. **Louise H. Comas:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Leonardo Deiss:** Writing – review & editing, Software, Formal analysis. **Tyler C. Donovan:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Meagan Schipanski:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Contributions

TD collected and analyzed the data and wrote the manuscript. All authors contributed to the study design, data interpretation, and reviewing and editing the manuscript.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agwat.2026.110456](https://doi.org/10.1016/j.agwat.2026.110456).

Data availability

Data will be made available on request.

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Glossary

- Agronomic optimum nitrogen rate:** The nitrogen availability at which predicted grain yield (or other response variable) is at its maximum
- Evapotranspiration (ET_c act):** Actual crop evapotranspiration representing total crop water use (soil evaporation + plant transpiration) using soil water balance during the growing season.
- Water Productivity:** Grain yield per unit of crop evapotranspiration ($\text{kg grain mm}^{-1} \text{ha}^{-1}$)
- Nitrogen Use Efficiency (NUE):** General term describing crop nitrogen performance metrics related to grain yield, plant N uptake, and nitrogen inputs.
- Nitrogen Utilization Efficiency (NUE):** Grain yield per unit of total plant N uptake
- Partial Nitrogen Balance Grain (PNB_{grain}):** Plant N uptake per unit of N applied
- Nitrogen Availability:** Total nitrogen ($\text{NH}_4\text{-N} + \text{NO}_3\text{-N}$) supply to the crop defined as the sum of fertilizer N, N in the irrigation water, and pre-plant residual soil N from 0 to 90 cm